

Course Project Requirements

- each group will identify a problem involving real data
- formulate an optimization problem
- apply optimization techniques on the problem

group size is 2-4

Project Milestones

1. Project Proposal and Motivation (10%)
5-minute presentations to teaching assistants in mid September
2. Mathematical Formulation (10%)
3. Solution (5%) and Analysis (5%)
10-15 minute presentations in mid November

Project Proposal, Motivation (10%)

1. introduce the group members and the chosen real-world data
2. describe the dataset: its source, size (e.g., number of data points, features), and what the key variables represent
3. Explain the real-world problem. Emphasis is on optimization. Answer the question: why do we need optimization? What are the impacts of solving the problem?

5-minute presentations to teaching assistants in mid September

Mathematical Formulation (10%)

1. formulate two problems on the same data: 'easy' and 'hard'; elucidate the variables, objective function, and any constraints in mathematical form
2. 'easy' problem can be a linear program or an unconstrained convex problem that solves the original problem approximately
3. 'hard' problem should add a layer of complexity to the easy one and more faithfully capture the original problem; examples include non-convex objectives, non-linear constraints

Solution (5%) and Analysis (5%)

1. implement and solve; recommended software is python; verification of the optimal solutions is needed; for instance, report the final objective value and values of key variables (5%)
2. compare 'easy' and 'hard' problems. How did the added complexity change the optimal solution? What is the 'price of realism' (i.e., how much did the optimal value change)? Visualize and discuss the key trade-offs between easy, hard (5%)

10-15 minute presentations in mid November

Option 1: General / Machine Learning Datasets

- regression, classification, feature selection, ...
- loss minimisation, regularisation, sparse modelling, ...
- easy → hard: convex regression/logistic model → feature selection, fairness or robustness constraints

<https://www.kaggle.com/datasets>

<https://huggingface.co/datasets>

<https://archive.ics.uci.edu/>



Option 2: Transport, Routing & Logistics

- road networks, mobility demand, delivery, ...
- shortest path, vehicle routing, fleet positioning, ...
- easy → hard: transportation LP → time windows, vehicle capacity and integer routing decisions



<https://www.openstreetmap.org/>

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

<https://gtfs.org/resources/sharing-data/>

Option 3: Healthcare & Biomedical Data

- clinical, physiological, diagnosis, ...
- classification loss minimization, sparse prediction, resource allocation, ...
- easy → hard: : regularized logistic regression → interpretability, fairness or clinical constraints



<https://physionet.org/data/>

<https://archive.ics.uci.edu/>

<https://www.kaggle.com/datasets?search=healthcare>

Option 4: Agriculture, Climate & Environment

- crop production, weather, climate, development and sustainability data, ...
- crop allocation, water planning, renewable siting, food distribution, ...
- easy → hard: linear land/water allocation → nonlinear yield, rotation and uncertainty constraints



<https://www.fao.org/faostat/en/>

<https://www.ncei.noaa.gov/cdo-web/>

<https://data.worldbank.org/>

Option 5: Networks & Graph Optimization

- social, citation, road, web, collaboration and biological networks, ...
- max-flow/min-cut, graph partitioning, influence maximization, matching, ...
- easy $\rightarrow$ hard: flow or cut formulation $\rightarrow$ integer decisions, budgets and nonlinear influence models



<https://snap.stanford.edu/data/>

<https://networkrepository.com/>

<https://konect.cc/networks/>

Option 6: Finance & Economics

- market data, macroeconomic indicators and portfolio datasets, ...
- portfolio optimization, risk-return trade-offs, hedging and allocation, ...
- easy $\rightarrow$ hard: convex mean-variance QP $\rightarrow$ transaction costs and cardinality constraints



<https://fred.stlouisfed.org/>

<https://finance.yahoo.com/>

<https://people.brunel.ac.uk/~mastijb/jeb/orlib/portinfo.html>

Option 7: Operations Research Benchmarks

- ready-made optimization instances with natural objectives and constraints, ...
- VRP, assignment, scheduling, bin packing, facility location, set covering, ...
- easy → hard: LP relaxation or simplified model → full integer/nonlinear benchmark instance



<https://people.brunel.ac.uk/~mastijb/jeb/info.html>

<https://miplib.zib.de/>

<https://plato.asu.edu/ftp/lptestset/>

Option 8: Any Other Domain that Excites You

e.g., energy and
electrical systems



<https://datasetsearch.research.google.com/>

Final Report + Code Submission

1. submit final Report and code on LMS in **early December**
2. Generative AI can be used at all levels (report, code, slides)
your role is **verifying and organising generated content**
3. project will be assessed based on the presentations and the final report + code